

PSET 1: Notation, sets and functions, quadratics, and logarithms

Assignment Qs

- Name
- How long did this problem set take you? 0-1 hour 1-2 hours 3-5 hours 5+ hours
- How difficult was this problem set? very easy 1 2 3 4 5 very challenging

Problem 1: Translate a statement into notation

A researcher proposes a simple model of protest participation: an individual joins a protest when the benefit they expect from the protest succeeding, weighted by the chance their participation matters, exceeds the personal cost of participating (time, risk of arrest, etc.).

- Define notation for each quantity in the model (in the style of $R = PB - C$ from the rational voter model in class) and write the condition under which an individual participates.
- Using your notation, write the formal statement of the following implication: “in a very large protest movement, any one person’s participation is unlikely to matter, so individuals should stay home.”
- Suppose the government announces harsher penalties for protesters. Which term in your model changes, in which direction, and what does the model predict about participation?

Problem 2: Work with sets

Using the sets¹

$$\begin{aligned}A &= \{2, 3, 7, 9, 12, 13\} \\B &= \{x : 6 \leq x \leq 10 \text{ and } x \text{ is an even integer}\} \\C &= \{x : 2 < x < 20 \text{ and } x \text{ is prime}\} \\D &= \{1, 4, 9, 16, 25, \dots\} \\E &= \{1, 4\}\end{aligned}$$

identify the following:

- $A \cup B$
- $(A \cup B) \cap C$
- $C \cap D$
- $A \times E$

¹inspired by Grimmer HW1.1

Problem 3: Functions vs. relations

- a. True or false—explain why: “All relations are functions, but not all functions are relations.”
- b. For each of the following, state whether y is a function of x (Y/N). If it is a function, state its domain. If it is not, state a restriction that would make it one.
- i. $y = x^2$
 - ii. $x = y^2$
 - iii. $y = x^{1/2}$
- c. Is $y = x^2$ invertible on all of $\mathbb{R}$? If not, give a domain on which it is invertible.

Problem 4: Simplify expressions

Simplify the following expressions as much as possible:²

- a. $(-x^2y^3)^3$
- b. $(-2)^{(4-9)}$
- c. $\left(\frac{1}{27b^4}\right)^{1/3}$
- d. $\frac{13a/7b}{13b/2a}$
- e. $(a+b)^2 + (a-b)^2 + 2(a+b)(a-b) - 3a^2$

Problem 5: Graph sketching

Let the functions $f(x)$ and $g(x)$ be defined for all $x \in \mathbb{R}$ by³

$$f(x) = \begin{cases} |x-2| & \text{if } x < 1 \\ 1 & \text{if } x \geq 1 \end{cases}, \quad g(x) = \begin{cases} x^3 & \text{if } x < 2 \\ 4 & \text{if } x \geq 2 \end{cases}$$

- a. Sketch $y = f(x)$
- b. Compute $f(g(0))$ and $f(g(3))$
- c. Sketch $y = f(g(x))$
- d. Sketch $y = g(f(x))$

²inspired by Gill 1.1

³inspired by Pemberton and Rau problem 3-1

Problem 6: Root finding

Find the roots (solutions) of the following quadratic equations.⁴ Recall that for $ax^2 + bx + c = 0$,

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

a. $9x^2 - 3x - 12 = 0$

b. $x^2 - 2x - 16 = 0$

c. $6x^2 - 6x - 6 = 0$

Problem 7: Simplify logarithms

Express each of the following as a single logarithm:⁵

a. $\log(x) - 2\log(y) + \log(z)$

b. $2\log(x) + \log(1)$

c. $\log(2x) - 2$

AI and Resources statement

Please list (in detail) all resources you used for this assignment. If you worked with people, list them here as well. It is not enough to say that you used a resource for help, you need to be specific on the link and *how* it was helpful. W/R/T gen AI tools (including GPT, Claude, etc.) you cannot use them to do work on your behalf—you cannot put in any of the questions, etc. You can ask for help on logic / sample problems. If you do use GPT or other AI tools, you need to provide a link to your chat transcript. Any suspected academic integrity violations will be immediately reported to the Dean of Students.

⁴Gill 1.25

⁵Inspired by Grimmer HW1.3. Assume log is natural log unless specified otherwise.